PRIYANSHU PRAKASH

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SUMMARY

Data Scientist,ML Engineer skilled in building high-impact predictive models, optimizing pipelines, and delivering measurable accuracy gains.Experienced in NLP, LLMs, and end-to-end machine learning systems that convert raw data into actionable insights.Strong engineering mindset with proven success improving model performance, automating workflows, and deploying scalable AI solutions.

EDUCATION

Vellore Institute of Technology, Vellore — **Graduated August 2025**
Integrated M.Tech in Computer Science (Specialization in Data Science)

Delhi Public School, Ranchi — **July 2020**
Class XII – CBSE

Delhi Public School, Ranchi — **July 2018**
Class X – CBSE

ACADEMIC PROJECTS

GenAI-Powered Cold Email Generator for Business Development — *July 2025*
Python • LangChain • Llama 3.1 (Groq) • ChromaDB • Streamlit

- Built and deployed a **fully automated LLM-driven email generator** that crafts personalized outreach emails based on real-time job postings.
- Implemented **LLM prompt chaining + semantic vector search**, improving skill extraction accuracy by **30% compared to baseline keyword extraction**.
- Engineered embeddings pipeline using ChromaDB, enabling **sub-100ms retrieval latency** and **92% match relevance** against test job postings.
- Developed a Streamlit UI used by non-technical stakeholders, reducing email creation time **from 15 minutes to** $<$**20 seconds**.

Multi-Objective Recommendation System using Evolutionary Algorithms — *Sep 2024 – March 2025*
Python • NSGA-II • Scikit-learn

- Designed a multi-objective recommender system optimizing **accuracy, diversity, and novelty** using NSGA-II.
- Achieved a **26% improvement in recommendation diversity** with **no accuracy drop**, outperforming baseline evolutionary algorithms.
- Built custom evaluation metrics and compared across multiple datasets.

Crime Hotspot Detection & Temporal Crime Prediction — *July 2024*
Python • K-Means • SARIMA • Pandas • Matplotlib

- Identified crime hotspots across Telangana using geo-spatial clustering; improved cluster separation score by **18%** through targeted feature engineering.
- Developed SARIMA-based forecasting models achieving **85% prediction accuracy**, enabling early visibility into rising crime trends.

Brain Tumor Prediction Using Machine Learning — *Dec 2023*
Python • Scikit-learn • EDA

- Analyzed **570 patient records** to identify key tumor predictors and correlations.
- Built ML models achieving **93% classification accuracy** and a **0.91 F1-score**, outperforming logistic regression baselines by **14%**.
- Visualized high-impact features to improve interpretability for clinical decision-making.

Airbnb Price Prediction
Nov 2023

Python • Pandas • Scikit-learn • TensorFlow/Keras • Regression • ANN • EDA • Feature Engineering

- Analyzed a dataset of **50,000 Airbnb listings** to predict rental prices using location features, amenities, host verification, and property characteristics.
- Performed comprehensive **Exploratory Data Analysis (EDA)** to identify key predictors and correlations between features and the target price, improving feature relevance.
- Engineered features such as amenity counts, neighborhood-level averages, property-type encodings, and host verification signals to enhance model signal quality.
- Developed and compared multiple models—including **Linear Regression, Random Forest Regressor, and Artificial Neural Networks (ANNs)**—achieving a **17–22% reduction in RMSE** over baseline models.

Experience

MECON LIMITED — Vocational Training *Ranchi, India* *(Jun 2023 – Jul 2023)*

Audio Classification • Feature Engineering

- Developed an audio classification model to identify urban environmental sounds using MFCC-based acoustic features.
- Improved model accuracy by **22%** through optimized feature extraction and targeted parameter tuning.

Skills

Languages: Python (NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn)
Machine Learning: Supervised Learning, Unsupervised Learning, Feature Engineering, Model Optimization, Time-Series Forecasting
NLP: Tokenization, Named Entity Recognition (NER), Summarization, Embeddings, Vector Search
Generative AI: Llama 3.1 (Groq), Hugging Face Transformers, LangChain, RAG Pipelines, ChromaDB, Prompt Engineering
Tools & Platforms: SQL, Git/GitHub, Streamlit, Hadoop, Tableau
Certifications: IBM — Data Science Methodology; Udemy — ML A–Z; TechBrain — ML Using Python